

SAKETH PAPAREDDY

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SUMMARY

AI Graduate Student (MS AI, UNT) with hands-on experience building end-to-end machine learning systems. Developed a FAISS-based retrieval pipeline with cross-encoder reranking (17× latency reduction), a dual-model sales forecasting system using FastAPI and Flask, and an XGBoost-based recommendation engine (NDCG@5: 0.927). Focused on building deployable, production-oriented ML solutions.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Pandas, NumPy, Matplotlib, Scikit-learn

ML / Deep Learning: TensorFlow, PyTorch, XGBoost, CNNs, RNNs, LSTMs, ARIMA, Feature Engineering

Generative AI & LLMs: LangChain, FAISS, BM25, Sentence Transformers, Groq API, Llama 3.3 70B, Mistral-7B (4-bit)

Deployment & Tools: FastAPI, Flask, Streamlit, Git, GitHub, Jupyter, Apache Spark, Hadoop

Cloud & Certifications: Microsoft Azure (AZ-900 certified)

PROJECTS

Generative RAG Pipeline with Hybrid Retrieval & LLM Generation *Python, LangChain, FAISS, BM25, Groq, Llama 3.3 70B, Streamlit*

- Built and deployed a PDF Q&A system on Streamlit Cloud using Llama 3.3 70B (Groq API) with Mistral-7B local fallback, cutting response time from ~45s to 2.6s (17x faster).
- Implemented hybrid retrieval (FAISS + BM25 + cross-encoder reranking) and a 6-class query router — 90% classification accuracy, 100% out-of-scope rejection.
- Validated against 48 questions on a 176-page technical document; achieved F1=0.51 with hallucination filtering via n-gram overlap grounding.

Retail Store Sales Forecasting Engine *Python, ARIMA, LSTM, TensorFlow, FastAPI, Flask, Pandas*

- Built and served two forecasting models — ARIMA (RMSE: 0.0722, directional accuracy: 78.57%) via FastAPI REST API and LSTM (RMSLE: 0.2892, directional accuracy: 67.48%) via Flask web interface — both accessible without writing code.
- Built a feature pipeline across 5 external signals (oil prices, holidays, promotions, store metadata, transactions) with N-step forecasting and CSV export endpoints.

Airbnb Destination Recommendation System *Python, XGBoost, Gradient Boosting, Flask, Pandas, Scikit-learn*

- Trained XGBoost on 213K+ user records to rank top-5 destinations (NDCG@5: 0.927), served via a Flask web app, and ranked top 15% on Kaggle.
- Engineered 500+ features from session logs, demographics, and device signals; trained a second Gradient Boosting model (NDCG@5: 0.824) on booking-only users to compare behavior across user segments.

EDUCATION

University of North Texas

Denton, TX

Master of Science in Artificial Intelligence

Expected Graduation: May 2026 | Relevant Coursework: Deep Learning, Machine Learning, Neural Networks, Computer Vision

NBKR Institute of Science & Technology

Vidyanagar, A.P., India

Bachelor of Technology in Information Technology

LEADERSHIP & CERTIFICATIONS

- Directed 'Inforium', a department-wide technical event at NBKRIST — coordinated 10+ volunteers, owned vendor logistics, and delivered to 200+ attendees on time.
- Microsoft Certified: Azure Fundamentals (AZ-900)